IoT-Based Area Measurement and Image-Based Crack Detection for Cement Cube Compressive Strength Testing

25-26J-250

Project Proposal Report

Lakshan V.G.S

B.Sc. (Hons) Degree in Information Technology

Specializing In Information Technology

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Declaration

I declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration. The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor: Date: 2025/08/29

Signature of the co-supervisor: Date: 2025/08/29

ABSTRACT

IoT-Based Area Measurement and Image-Based Crack Detection for Cement Cube Compressive Strength Testing

The primary objective of this project is to develop an automated system for evaluating the compressive strength of cement cubes by integrating non-contact surface measurement and image-based crack detection. The system is designed to replace manual vernier caliper measurements and subjective crack inspection with a reliable, sensor- and AI-driven solution that improves the accuracy and efficiency of laboratory testing.

At present, cement cube testing relies heavily on vernier calipers to measure cube dimensions and on visual judgment to identify the first crack under compressive loading. These methods are labor-intensive, prone to human error, and provide only limited real-time feedback. In addition, destructive testing procedures delay results, while manual observation of crack propagation is inconsistent and may fail to detect early-stage microcracks. Such limitations reduce the reliability of quality control in concrete testing and can impact structural safety.

To address these issues, this project proposes a system that combines distance sensors such as ultrasonic and infrared devices for cube face dimension measurement with multi-camera image acquisition and deep learning algorithms for crack detection. Surface images will be processed using digital image processing techniques and convolutional neural networks (CNNs) to identify crack initiation with high precision. At the same time, force data collected from load cells will be synchronized with the moment of crack detection, enabling real-time calculation of compressive strength at failure. The system will also provide an alert mechanism to notify operators when a crack forms and record critical parameters such as length, width, and surface area of the cube face.

The proposed system targets construction quality control laboratories, cement manufacturing companies, and civil engineering researchers who require accurate, rapid, and cost-effective strength assessment of concrete cubes. It draws on multiple technological components, including ultrasonic and infrared sensors for length and width measurement, computer vision through a multi-camera setup, convolutional neural networks such as InceptionV3 for automated crack identification, digital image processing for noise reduction and edge detection, and concepts from structural health monitoring that incorporate electromagnetic and microwave sensor technologies.

The expected outcome of this project is an automated platform that can deliver accurate area measurement of cube faces without manual tools, detect cracks at an early stage with high precision, and calculate compressive strength in real time. By minimizing reliance on human judgment, the system enhances safety, repeatability, and efficiency in concrete testing. Furthermore, it contributes to sustainable construction practices by reducing the risk of overusing cement due to inaccurate testing and by ensuring reliable evaluation of concrete quality for structural applications.

Keywords: Concrete cube testing, Compressive strength, Crack detection, Distance sensors, Image processing, Convolutional Neural Networks, Non-destructive testing, Structural health monitoring, Civil engineering automation.

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LIST OF ABREVATIONS

Table I : List of Abbreviations

Abbreviation	**Full Form**
IoT	Internet of Things
CNN	Convolutional Neural Network
AI	Artificial Intelligence
ML	Machine Learning
NDT	Non-Destructive Testing
API	Application Programming Interface
IDE	Integrated Development Environment
MPa	Mega Pascal
LKR	Sri Lankan Rupee
DB	Database
SDLC	Software Development Life Cycle
UI	User Interface
UX	User Experience
USB	Universal Serial Bus
API	Application Programming Interface

1. INTRODUCTION

Concrete compressive strength testing is one of the most critical procedures in construction quality control, as it ensures the safety and durability of structures. Traditionally, cement cube testing relies on vernier calipers for dimension measurement and manual observation to detect cracks during compressive loading. These methods are time-consuming, labor-intensive, and prone to human error, particularly when microcracks appear before visible failure. The limitations of such approaches have motivated researchers to propose alternative solutions that employ digital image processing and artificial intelligence for non-destructive evaluation, providing faster and more reliable results (Doğan et al., 2024)

Recent studies demonstrate the potential of machine learning and computer vision for analyzing cube surfaces and predicting compressive strength. For example, digital image processing combined with artificial neural networks has been shown to achieve high accuracy in estimating concrete strength from surface images (Doğan et al., 2024), while CNN-based models such as InceptionV3 have proven highly effective in detecting cracks in laboratory-scale concrete cubes with accuracies exceeding 97%(Kapadia et al., 2023). Similarly, Industry 4.0-inspired approaches integrating image processing with real-time monitoring highlight the advantages of automated crack detection and damage classification(Patil et al., 2023). These findings emphasize that intelligent systems can significantly enhance the reliability of compressive strength evaluation compared to traditional manual testing.

Despite these advancements, challenges remain. Image-based systems are sensitive to noise, lighting variations, and uneven cube surfaces, while sensor-based methods can be influenced by environmental factors and require careful calibration. Economic considerations also play a role, as advanced multi-sensor systems can be costly compared to conventional testing tools. To overcome these issues, this research proposes a hybrid solution that combines distance sensors for accurate length and width measurement(Kassim et al., 2013)with multi-camera vision systems for automated crack detection. By synchronizing force data with real-time image analysis, the system will calculate compressive strength at the precise onset of cracking. The expected outcome is a robust, cost-effective, and sustainable platform that improves accuracy, reduces human dependency, and supports safer construction practices(Patil et al., 2023)(Gkantou et al., 2019)

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1.1 BACKGROUND AND LITERATURE SURVEY

Accurate measurement of concrete cube dimensions and precise evaluation of compressive strength are fundamental components of modern construction quality control. The accuracy with which cube surfaces are measured and cracks are detected directly impacts the reliability and safety of structures. Historically, these tasks have been carried out using vernier calipers for length and width measurement and manual visual observation for crack detection during compressive testing. While effective in certain contexts, these traditional approaches are time-consuming, labor-intensive, and prone to human error, particularly in the early identification of microcracks that are invisible to the naked eye.(Kapadia et al., 2023)

With the advent of sensor technology and computer vision, new approaches have emerged for non-contact measurement and automated crack detection. Distance sensors such as ultrasonic and infrared devices allow for more streamlined and contact-free surface dimension measurement, eliminating the limitations of manual calipers. However, infrared sensors are sensitive to environmental conditions and object properties, making ultrasonic sensors a more stable and accurate option for industrial applications.(Kassim et al., 2013) Meanwhile, advances in digital image processing and machine learning have enabled automated crack detection with high precision. For example, convolutional neural networks (CNNs) have been used to classify and analyze cracks on concrete cubes with accuracies exceeding 97%, offering reliable real-time crack monitoring compared to traditional methods.(Kapadia et al., 2023)

The integration of sensors and AI-driven image analysis has opened new possibilities for improving the accuracy, efficiency, and repeatability of cube testing. Yet, the full potential of these advancements is not fully realized, particularly in terms of combining synchronized force measurement, dimension sensing, and crack detection into a single, automated platform. This research proposal aims to develop such a system, bridging the gap between conventional destructive testing and intelligent, non-contact, automated solutions.

The research paper *"Convolutional Neural Network Based Improved Crack Detection in Concrete Cubes"* demonstrates the potential of CNNs such as InceptionV3 for identifying cracks in laboratory-scale cubes. With a dataset of over 160,000 images, the model achieved validation accuracies above 97% and an F-score of 0.93, proving the feasibility of real-time automated crack detection (Kapadia et al., 2023). Similarly, Patil et al. (2024) proposed an inspired approach combining image processing and CNNs for cube quality testing. Their study highlighted that automated crack detection not only improves reliability but also reduces cement overuse caused by misclassification of failure modes, thereby supporting sustainable practices in construction(Patil et al., 2023).

Sensor-based approaches also provide critical support for dimension measurement. Kassim et al. conducted a comparative study on ultrasonic and infrared distance sensors, concluding that ultrasonic sensors maintain accuracy and stability across diverse materials and environments, where infrared sensors often fail. (Kassim et al., 2013)This finding is highly relevant for non-contact cube measurement, as cube faces are rarely uniform and can exhibit reflective or uneven surfaces. Complementary to this, electromagnetic and microwave sensors have been investigated for structural health monitoring, showing strong correlations between crack propagation and signal response under loading, suggesting additional opportunities for embedded sensing in concrete elements.(Gkantou et al., 2019)

Beyond laboratory systems, IoT-based prototypes demonstrate the feasibility of low-cost, connected crack monitoring platforms. Recent implementations using ESP32-CAM modules with ultrasonic sensors and alert systems illustrate how portable, wireless systems can provide real-time data capture and analysis. Meanwhile, studies on image-based compressive strength prediction reveal that AI models can achieve high accuracy when trained on surface images, though performance tends to decrease when transitioning from controlled lab samples to real-world data, highlighting the importance of robust datasets.(Doğan et al., 2024)

Taken together, these studies show that while significant progress has been made in both non-contact measurement and image-based crack detection, there remains a need for an integrated system that unifies sensor-based surface area measurement, AI-powered crack detection, and synchronized force monitoring into a single automated platform. This proposed research aims to fill that gap by designing a hybrid system capable of delivering accurate, real-time compressive strength evaluation and early crack detection with operator alerts and digital reporting.

2. RESEARCH GAPS

The effectiveness of compressive strength testing in cement cubes has significantly improved with the introduction of image processing, convolutional neural networks (CNNs), and sensor-based measurement techniques. These technologies provide faster; more objective evaluations compared to traditional methods such as vernier calipers and visual crack observation. However, despite these advancements, several critical limitations remain. Image-based systems are highly sensitive to lighting conditions, noise, and surface irregularities, while sensor-based methods such as infrared measurement can fail under certain material properties.

Because of these issues, there are substantial challenges facing construction companies and quality control laboratories. Accurate measurement of cube dimensions and early crack detection are essential for determining compressive strength reliably. Current tools often fall short of this requirement, as manual methods are inconsistent, and AI-driven models trained in laboratory conditions may lose accuracy when applied in real-world environments. This lack of robustness can lead to inaccurate classification of strength, material waste, and reduced safety margins in structural applications.

In addition, while several studies have explored either sensor-based dimension measurement or CNN-based crack detection, few have attempted to integrate these into a single automated framework. Current research tends to address one problem in isolation, leaving a gap in systems that simultaneously capture cube dimensions, synchronize applied force, and detect cracks in real time. This integration is critical for providing automated alerts, accurate $\sigma = F/A$ calculations, and digital reporting. This gap in research and technology highlights the need for a comprehensive solution that combines dimension sensing, force monitoring, and machine learning-based crack detection into a single, user-friendly platform. By addressing this gap, it becomes possible to create a system that not only improves the reliability of cube strength testing but also reduces human error, saves time, and supports sustainable construction practices.

Table II: Research Gap

Conducted By	Accurate Measurement	Automated Crack Detection	Real-Time Force Integration	Comprehensive Solution
Research A(Kapadia et al., 2023)		✓		
Research B (Patil et al., 2023)		✓		
Research C (Kassim et al., 2013)	✓			
Research D (Gkantou et al., 2019)		✓		
Research E (Doğan et al., 2024)	✓	✓		
Proposed System	✓	✓	✓	✓

3. RESEARCH PROBLEM

The Sri Lankan cement industry plays a crucial role in supporting national infrastructure development, yet one of the persistent challenges is the accurate and timely prediction of cement compressive strength. Conventional laboratory testing methods, which are performed post-production, are slow, resource-intensive, and reactive. This delay in obtaining results hinders quality assurance processes and prevents manufacturers from making real-time adjustments during production. As a result, there is an increased risk of producing cement batches that fail to meet required standards, leading to wasted resources, financial losses, and compromised structural reliability.

The core problem lies in how to enhance the accuracy and efficiency of compressive strength prediction using modern technological solutions. Current practices face limitations in integrating real-time chemical, physical, and process data, as well as adjusting for environmental factors such as temperature and humidity. Moreover, the absence of early warning systems and batch-specific recommendations restricts manufacturers from proactively optimizing processes, which in turn increases operational costs and reduces competitiveness in the industry.

To address these issues, there is a need for an advanced, data-driven approach that leverages machine learning and predictive analytics. Such a system should be capable of processing multi-source production data, learning from laboratory outcomes, and delivering instant strength predictions at multiple curing ages. Additionally, it should identify key influencing parameters, provide actionable feedback for quality control, and generate early warnings for potential deviations. By introducing this predictive capability, Sri Lankan cement manufacturers will be able to ensure consistent product reliability, reduce waste, lower costs, and achieve greater operational efficiency.

4. OBJECTIVES

4.1 MAIN OBJECTIVES

The primary aim of this research is to develop an automated, intelligent system that will assist construction laboratories and cement companies in performing more accurate and reliable compressive strength testing of cement cubes. The system will focus on three main areas: accurately measuring cube dimensions using non-contact sensors, detecting crack initiation in real time through image processing and machine learning, and automatically calculating compressive strength by integrating force and area data. The goal is to provide engineers and technicians with a user-friendly platform that reduces human error, improves testing efficiency, and ensures safer and more sustainable construction practices.

4.2 SPECIFIC OBJECTIVES

4.2.1 ACCURATE AREA MEASUREMENT

One of the primary objectives of this research is to develop a system that provides accurate and reliable cube surface area measurement capabilities. The system will allow technicians to measure the length and width of the cube faces using non-contact sensors such as ultrasonic devices, eliminating the errors commonly associated with manual vernier caliper methods. By integrating advanced sensor technology with digital calibration, the tool will ensure that surface area calculations are precise, overcoming the inaccuracies of traditional approaches. This will provide laboratories with dependable data, which is essential for accurate compressive strength evaluation. Additionally, the system will be designed to automatically record and display measurements, enhancing both accuracy and usability in testing environments.

4.2.2 INTERACTIVE FIELD MANAGEMENT TOOLS

One of the primary objectives of this research is to develop a system that ensures both accurate cube surface area measurement and real-time crack detection during compressive strength testing. The system will use non-contact sensors, such as ultrasonic devices, to precisely measure the length and width of cube faces, eliminating the errors and inconsistencies of traditional vernier caliper methods. At the same time, a multi-camera setup integrated with image processing and machine learning techniques will continuously monitor the cube surface to identify crack initiation with high precision, overcoming the subjectivity and delays of manual visual observation. By combining reliable surface area measurements with automated crack detection, the system will provide laboratories with dependable data for accurate compressive strength calculation, while also offering a more efficient, user-friendly, and objective testing process.

4.2.3 ECONOMIC ANALYSIS AND COST MANAGEMENT

To help construction laboratories and cement companies achieve more efficient resource management, the proposed system incorporates capabilities for automated economic analysis within the testing process. With the help of these real-time analytical tools, engineers will be able to plan material usage, evaluate testing costs, and make informed decisions about quality control methods. Using information derived from cube dimension measurements, applied load, and detected cracks, the tool will ensure that the economic analysis is accurate, timely, and relevant to industry needs. This will enable organizations to allocate resources more effectively, reduce unnecessary expenditures caused by inaccurate manual testing, and ultimately increase overall efficiency and profitability. The goal is to encourage sustainable construction practices by minimizing material wastage and improving cost-effectiveness. With these cost management features integrated into the automated cube testing system, laboratories will have a complete solution to address one of today's major challenges in construction quality assurance.

5. METHODOLOGY

5.1 SYSTEM OVERVIEW DIAGRAM

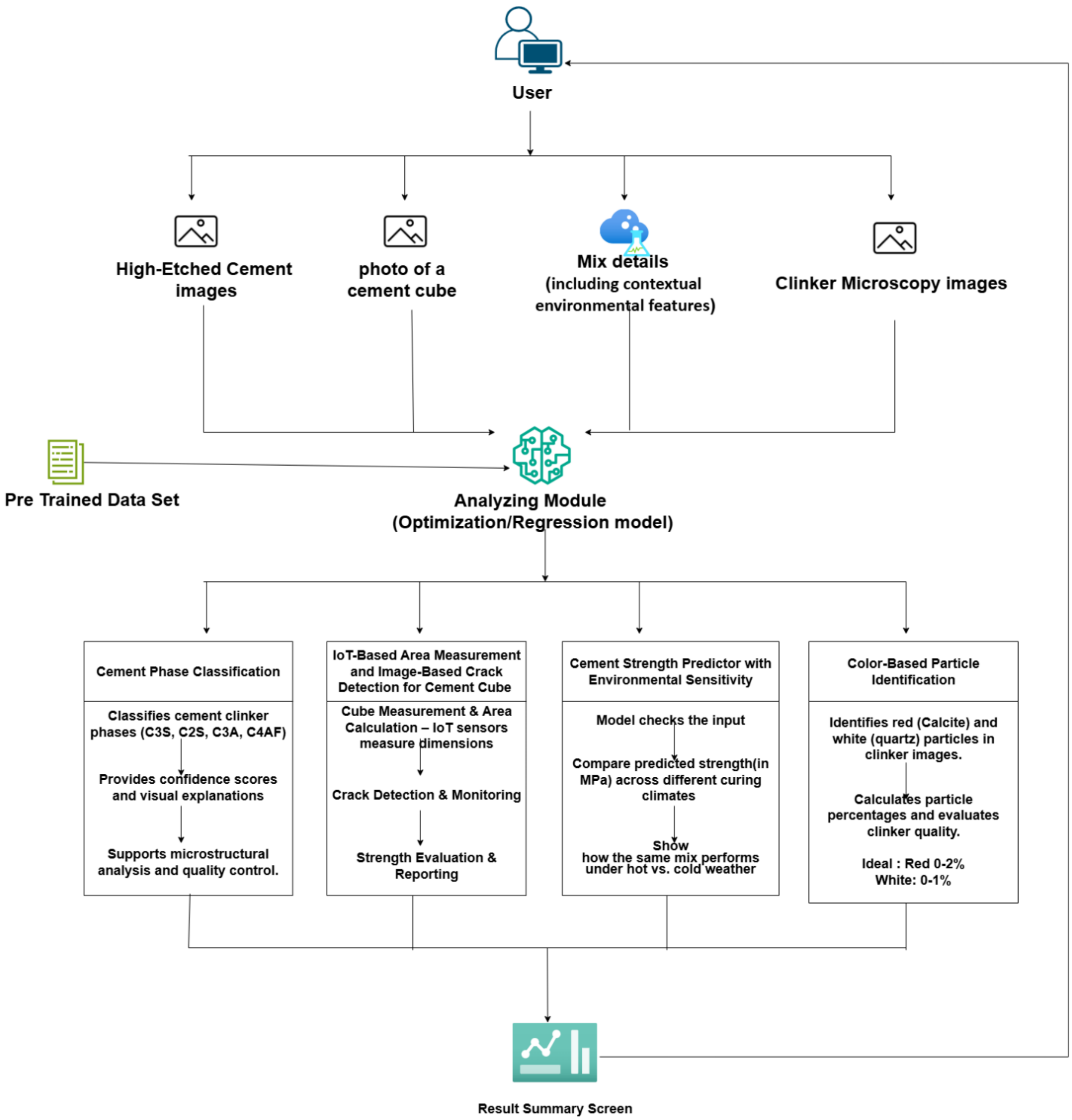

Figure 1 : System Overview Diagram

A comprehensive automated system is being developed as a solution to improve the accuracy, efficiency, and reliability of cement cube compressive strength testing. The system integrates sensors, cameras, and machine learning models with a user-friendly interface for laboratories and cement companies. The primary goal is to replace error-prone manual processes with real-time, automated solutions that ensure precise dimension measurement, accurate crack detection, and instant compressive strength calculation.

Dimension Measurement

The system incorporates non-contact ultrasonic sensors to measure the length and width of cement cube faces. Unlike traditional vernier calipers, which are prone to human error and inconsistencies, ultrasonic sensors provide accurate and repeatable measurements. The sensor readings are processed in real-time, and the calculated area of the cube face is stored automatically. To further enhance accuracy, the system applies digital calibration techniques to minimize noise and environmental effects.

Crack Detection

One of the core features of the system is real-time crack detection. A camera module is used to continuously monitor the cube during compressive loading. The images captured are processed through computer vision techniques using OpenCV and classified using a Convolutional Neural Network (CNN). This AI-based approach enables the system to detect microcracks that are not visible to the human eye, ensuring more reliable identification of the exact crack initiation point. The detected cracks are highlighted and logged with timestamps for further analysis.

Compressive Strength Calculation

The system calculates compressive strength by combining the measured cube surface area (from ultrasonic sensors) with the applied force values obtained from the compression testing machine. Using the formula:

$\sigma = F/A$

where σ is compressive strength, F is the applied force, and A is the cube surface area. This automated calculation removes manual errors and provides instant strength results at the exact moment of crack initiation.

Automated Alerts and Reporting

The system is designed to issue real-time alerts once cracks are detected or when strength thresholds are exceeded. Visual and audible notifications are provided to laboratory staff, ensuring timely response. Additionally, the system stores all measurement data, crack images, and strength calculations in a database. Reports can be generated in PDF and CSV formats, making it easy to review and archive results for quality assurance and compliance purposes.

AI-Driven Insights and Recommendations

Beyond crack detection, the CNN model is also trained to classify crack severity and provide early warnings about potential structural weaknesses. This feature enhances decision-making by offering insights into cube performance underload. Over time, the system's machine learning models improve with more data, making predictions more accurate and reliable.

Data Management and Cloud Integration

The system uses Firebase/MongoDB for efficient storage of test data and integrates with Azure Cloud services to ensure secure, scalable data management. This allows laboratories to store large volumes of test results, access them from multiple devices, and maintain backups. Cloud integration also enables remote monitoring and reporting, making the system highly adaptable for both academic research and industrial applications.

5.2 COMPONENT SYSTEM DIAGRAM

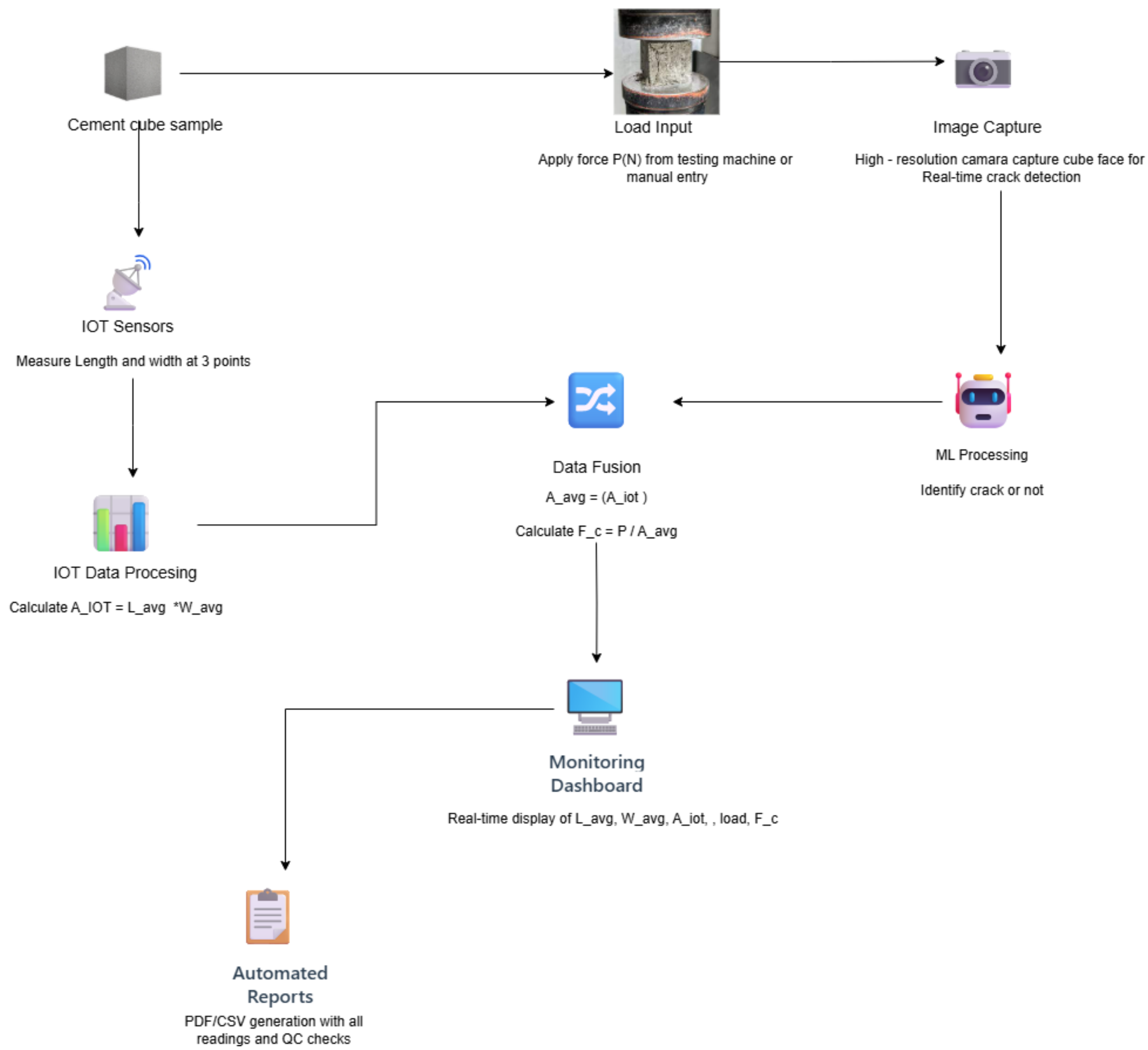

Figure 2 : Component System Diagram

According to the above diagram, users can test cement cubes efficiently through an IoT-enabled and AI-assisted quality assessment system. The system is designed to be user-friendly, ensuring that laboratory staff, engineers, and even students can operate it with minimal training. Its features enhance the accuracy, speed, and reliability of compressive strength testing, which is critical for quality assurance in construction.

Users begin by setting up a cement cube sample in the testing machine. IoT sensors then measure the cube's length and width at three different points, ensuring precision even when the cube surfaces are slightly irregular. These measurements are processed to calculate the average area of the cube face, which is used later for strength computation.

Simultaneously, the system allows load input either directly from the testing machine or by manual entry. At the same time, high-resolution cameras positioned around the cube continuously capture its surfaces. These images are analyzed in real-time using machine learning algorithms to detect cracks as soon as they begin to appear, ensuring no failure point goes unnoticed.

All sensor and image data are merged in the data fusion module, where the average cube area is finalized and the compressive strength is calculated using the formula fc=P/A_avg , f_c = P / A_{avg} f_c=P/A_avg. This calculation combines both IoT-based measurements and visual crack analysis, making the results more reliable than traditional single-method approaches.

The results are displayed on a real-time monitoring dashboard, which shows cube dimensions, load values, crack detection status, and computed strength. If a crack is detected or inconsistent measurements arise, the system automatically triggers an alert notification to the operator.

Finally, the system generates automated reports in PDF or CSV format. These reports include raw measurements, load data, crack detection images, and final strength values, making them suitable for documentation, quality audits, and research purposes.

This solution is a modern, integrated approach to cement testing, providing accuracy, early crack detection, real-time monitoring, and automated documentation. By combining IoT and AI technologies, it enhances safety, reduces manual error, and significantly improves the efficiency of compressive strength testing in the construction industry.

5.3 PROJECT TECHNOLOGY STACKS

Table III: Project Technology Stack

Tools & Technologies	Purpose
Python	For machine learning model development, image processing, and data analysis.
OpenCV	For image preprocessing and crack detection using computer vision.
TensorFlow / Keras	To train and deploy CNN-based machine learning models for crack detection.

Arduino / ESP32	For interfacing ultrasonic sensors and collecting non-contact length/width measurements.
Ultrasonic Sensor (HC-SR04 or equivalent)	To measure cube face dimensions (length and width) accurately.
Raspberry Pi / Jetson Nano	As the central processing unit for integrating sensors, cameras, and ML inference.
MongoDB / FireBase	For storing measurement data, crack detection logs, and test results.
Flask / Django	Backend framework to handle data processing and API services.
React (Web Dashboard)	To display real-time cube measurements, crack detection status, and compressive strength results in a user-friendly interface.
Matplotlib / Seaborn	For plotting force vs. time graphs and visualizing crack detection events.
Visual Studio Code	As the primary IDE for software development.
Git & GitHub	For version control and collaborative project development.
Postman	For testing backend APIs and system integration.

5.4 SOFTWARE SOLUTION

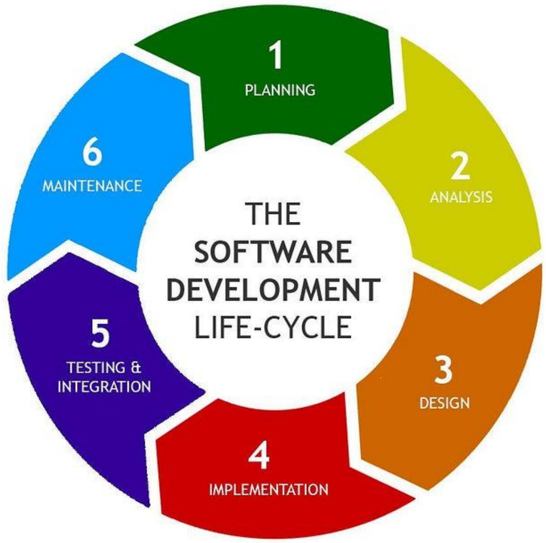

Figure 3 : SDLC Life Cycle

We will follow the Agile Software Development Life Cycle (SDLC) model to guide our project. Agile provides a flexible and iterative approach, allowing us to plan, develop, test, and deliver high-quality software in stages. This methodology encourages continuous improvement, frequent user feedback, and quick adjustments, ensuring the final product meets user needs effectively. By breaking the project into manageable sprints, the team can focus on delivering specific features incrementally, leading to a more efficient and responsive development process. This approach will help us build a robust and user-friendly land management application.

6. PROJECT REQURMENTS

6.1 FESIBILITY STUDIES

Conducting a feasibility study is essential to determine the ability to successfully develop and implement the proposed automated cement cube compressive strength testing system. The study evaluates the project from four key perspectives: technical, operational, economic, and schedule feasibility.

Technical Feasibility:
The project is technically feasible since it makes use of well-established and widely used technologies, including ultrasonic sensors for dimension measurement, load cells for force detection, and computer vision with CNN models (TensorFlow, OpenCV) for crack detection. The integration of these technologies using Python, Flask/Django, and React dashboards ensures that the system can be implemented effectively. The development team has the technical expertise to configure the hardware components, train machine learning models, and deploy the solution in a laboratory environment.

Operational Feasibility:
The proposed system is designed to meet the needs of construction laboratories and cement companies by providing accurate cube dimension measurement, real-time crack detection, and automated compressive strength calculation. The system reduces dependence on manual measurements and subjective crack observation, ensuring more reliable and repeatable test results. Feedback from engineers and technicians during prototype testing will guide refinements, ensuring that the system meets operational requirements and can be adapted into routine quality control workflows.

Economic Feasibility:
The project is economically viable, as the hardware components (e.g., ultrasonic sensors, load cells, cameras, microcontrollers) are affordable and easily available in the market. By reducing errors, preventing material wastage, and improving test efficiency, the system provides long-term cost savings for cement companies. The economic benefits outweigh the initial investment, making the solution cost-effective for laboratories seeking to modernize their testing processes.

Schedule Feasibility:
The project follows an Agile SDLC approach, enabling iterative development and integration of hardware and software components. This flexible model supports continuous improvement and allows timely identification of technical issues. A realistic project timeline has been set, with prototype development, testing, and validation scheduled to be completed within the academic year. The phased approach ensures that the system can be delivered on schedule and aligned with research milestones.

6.2 SURVEY RESULT

As part of this research, we conducted a field visit to INSEE Cement Company to collect practical insights and observations on the current processes used for cement cube compressive strength testing. The purpose of the field study was to understand the real challenges faced by technicians and engineers in measuring cube dimensions, detecting cracks, and calculating compressive strength during laboratory testing.

During the visit, we engaged directly with the technical staff, observed the cube testing procedures, and held informal discussions to gather their experiences and opinions. The focus was on identifying the limitations of manual methods, such as the use of vernier calipers for cube dimension measurement and visual observation for crack detection. The data collected during the visit was primarily qualitative, based on direct observation and staff feedback, but it provided valuable evidence of recurring issues in the testing process.

The findings from the visit indicated that inaccurate manual measurements, subjective crack detection, and the lack of integrated real-time calculation tools are major concerns. Technicians reported that early microcracks often go unnoticed, and calculations are delayed due to manual synchronization of load and area values. These limitations reduce the efficiency and consistency of test results. The observations gathered during the field visit are represented using summary tables and illustrative diagrams, which make the results easier to understand.

Based on these findings, we identified the need for an automated system that combines non-contact sensor-based cube measurement, AI-driven crack detection, and synchronized compressive strength calculation. These recommendations are directly aligned with the practical challenges observed at INSEE Cement Company and will ensure that the proposed solution is both relevant and industry ready.

6.3 USER REQUIREMENTS

- Enable accurate and non-contact measurement of cube surface area (length and width).
- Provide real-time crack detection during compressive strength testing using cameras and image processing.
- Integrate load cell data with surface area to automatically calculate compressive strength.
- Offer a user-friendly interface to display measurements, crack alerts, and strength results.
- Store and manage test results in a database for future reference and reporting.

6.4 FUNCTIONAL REQUIREMENTS

- Provide precise and non-contact cube dimension measurement using ultrasonic sensors.
- Enable real-time crack detection through camera input and image processing techniques.
- Integrate load cell data with measured cube surface area to automatically calculate compressive strength.

- Display results (dimensions, crack alerts, strength values) on a user-friendly web dashboard.
- Allow automatic data logging and storage of test results for reporting and future analysis.

6.5 NON-FUNCTIONAL REQUIREMENTS

- Usability: The system must have a simple and intuitive interface so that laboratory technicians and engineers, even with limited technical knowledge, can easily operate the cube testing system and interpret the results.
- Reliability: The system should operate consistently without failures during the testing process. It must securely handle data, provide accurate outputs, and give users confidence in the results.
- Performance: The system must deliver real-time and precise results for cube measurements, crack detection, and compressive strength calculations to ensure high performance and efficiency in laboratory operations.
- Compatibility: The system should be compatible with multiple platforms, including web-based dashboards accessible on Windows and macOS, and support integration with mobile devices for monitoring and reporting.

6.6 HARDWARE REQUIREMENTS

- Ultrasonic Sensors (HC-SR04 or equivalent): For accurate non-contact measurement of cube length and width.
- Camera Modules (ESP32-CAM / USB Cameras): For real-time image capture and crack detection.
- Raspberry Pi / Jetson Nano / Laptop or Desktop Computer: To process sensor data, run machine learning models, and integrate results.
- Stable Internet Connection (optional): For cloud storage, dashboard access, and report generation.

6.7 SOFTWARE REQUIREMENTS

- Desktop/Laptop: Windows 10/11, macOS, or Linux for development and execution.
- Embedded Platforms: Raspberry Pi OS (for Raspberry Pi) or Jetson Nano OS for hardware integration.
- Web Interface: Compatible with modern browsers (Chrome, Edge, Firefox) for accessing the testing dashboard.

6.7.1 BACKEND INFRASTRUCTURE

- Database: MongoDB / MySQL for storing cube measurements, crack detection logs, and compressive strength results.
- Backend Framework: Flask or Django (Python) for server-side logic, API handling, and integration of hardware data streams.
- Machine Learning Libraries: TensorFlow/Keras and OpenCV for training, deploying, and running crack detection models in real time.
- Data Visualization: Matplotlib / Seaborn for generating test reports and force–time graphs.

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6 7.2. DEVELOPMENT TOOLS

- IDEs: Visual Studio Code for coding, debugging, and integration.
- Version Control: Git and GitHub for source code management and collaboration.
- API Platform: Flask/Django REST Framework for building APIs that connect hardware with the web dashboard.
- Simulation & Testing: Python unittest / PyTest for validating system modules, and Postman for testing APIs.

6.8 TEST CASES

Table III : Test Cases

Test Scenario: Sensor Calibration

Test Scenario ID: T1

Description	Pre-Conditions	Inputs	Expected Results	Post-conditions
Verify ultrasonic sensors and cameras are calibrated before testing.	Hardware powered; cube placed; calibration targets/ArUco board available; app open.	Start 'Calibrate' → capture board → enter reference length; tare load cell.	Pixels-per-cm and sensor scale stored; load cell zeroed; status 'Calibration OK'.	Calibration profile saved; system ready for measurement.

Test Scenario: Length & Width Measurement (Non-contact)

Test Scenario ID: T2

Description	Pre-Conditions	Inputs	Expected Results	Post-conditions
Measure cube face L and W using ultrasonic (or vision scale).	Calibration completed (T1); cube face visible; sensors within range.	Press 'Measure' → system samples sensors (or detects edges).	L and W displayed within ±1 mm of reference; stability indicator green.	Measurements auto saved with timestamp and run ID.

Test Scenario: Area Calculation & Validation

Test Scenario ID: T3

Description	Pre-Conditions	Inputs	Expected Results	Post-conditions
Compute face area and validate against expected 150×150 mm cube.	T2 completed.	Click 'Compute Area'.	Area (A=L×W) computed; unit shown (cm^2); deviation vs nominal (%) displayed.	Area stored with L, W; ready for force sync.

Test Scenario: Real-Time Crack Detection (Vision)

Test Scenario ID: T4

Description	Pre-Conditions	Inputs	Expected Results	Post-conditions
Detect first crack during compression using camera(s) + ML.	Lighting on; cameras streaming 15–30 FPS; baseline captured.	Start 'Begin Test' → start press; system processes frames.	'Crack Detected' when confidence ≥ threshold; t_0 captured; snapshot saved.	Crack frame(s) and t_0 stored; proceed to strength calc.

Test Scenario: Force Acquisition & Time Sync

Test Scenario ID: T5

Description	Pre-Conditions	Inputs	Expected Results	Post-conditions
Stream load cell / machine force and align with video timestamps.	Load cell paired; sampling ≥10 Hz; common clock set.	Start force stream with test; optional sync flash.	Continuous F(t) with no gaps; latency <100 ms; sync marker recorded.	Force CSV saved; ready for $F(t_0)$ lookup.

Test Scenario: Compressive Strength at Crack Onset

Test Scenario ID: T6

Description	Pre-Conditions	Inputs	Expected Results	Post-conditions
Compute $\sigma crack=F(t_0)/A$ and display in MPa.	T3, T4, T5 completed; units configured.	Click 'Calculate Strength'.	Correct unit conversion; value shown with 2-decimal precision; uncertainty band displayed.	Result saved to run record; visible in dashboard.

Test Scenario: Alerts & UI Feedback

Test Scenario ID: T7

Description	Pre-Conditions	Inputs	Expected Results	Post-conditions
Verify audible/visual alert on crack and status indicators.	Speakers enabled; dashboard open.	Trigger crack (T4).	Buzzer + red banner; card shows L, W, A, $F(t_0)$, σ; operator guidance displayed.	Operator acknowledgement logged with user ID/time.

Test Scenario: Data Logging, Report Export

Test Scenario ID: T8

Description	Pre-Conditions	Inputs	Expected Results	Post-conditions
Save test artefacts and export PDF/CSV.	Completed test (T2–T7).	Click 'Export Report' & 'Download CSV'.	PDF includes metadata, L/W/A, force graph, crack images; CSV has time series.	Files stored under run folder; download links available.

7. DESCRIPTION OF PERSONAL AND FACILITIES

Table III : Description of personnel and facilities

Registration No	Name	Task Description
IT22562210	Lakshan V.G.S.	• Implement ultrasonic sensor integration for cube dimension measurement. • Configure load cell and amplifier for accurate force detection. • Set up multi-camera system for real-time crack monitoring. • Develop image preprocessing and CNN-based crack detection algorithms. • Synchronize force and area data to compute compressive strength automatically. • Design and implement backend for data storage and processing. • Develop web dashboard for displaying test results and alerts. • Conduct testing and validation at laboratory/industry partner facilities.

8. GANT CHART

Figure 4 : Gant Chart

9. WORK BREAKDOWN CHART

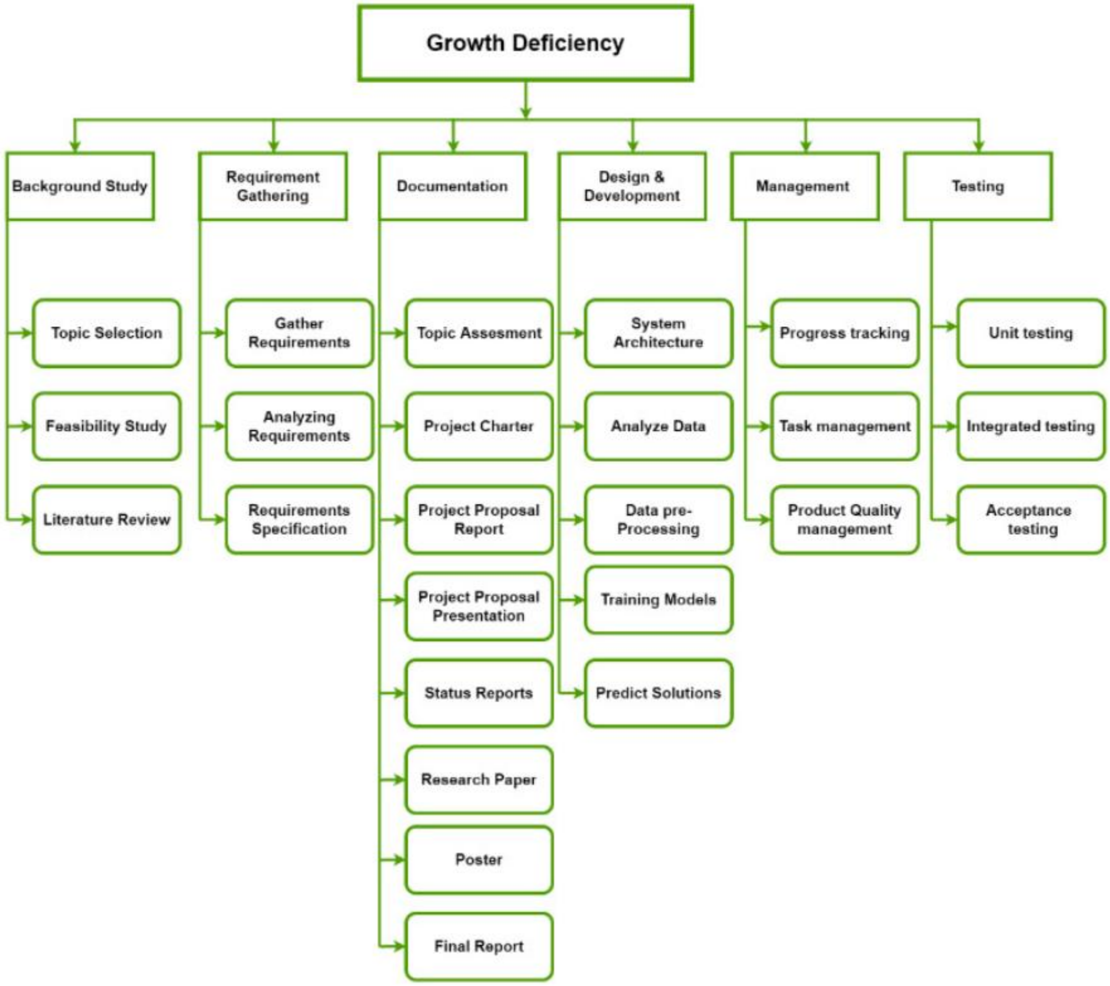


Figure 5 : WBS Diagram

10. BUDGET AND BUDGET JUSTIFICATION

Table IV : Budget and Budget Justification

Description	Amount (LKR)
Ultrasonic Sensors (HC-SR04 × 2)	5,000
Camera Modules (ESP32-CAM / USB × 2)	20,000
Raspberry Pi / Jetson Nano (1 unit)	45,000
Lighting & Fixtures (LEDs, mounts)	8,000
Data Storage & Database (MongoDB)	Free
Backend & Hosting (Flask/Django + Azure cloud)	30,000
Development Tools (VS Code, GitHub, Postman)	Free
Miscellaneous (Cables, mounts, shields, safety setup)	10,000
Total	**118,000**

11. CONCLUSION

In conclusion, this project aims to address the common problems encountered in traditional cement cube testing methods, particularly the reliance on manual caliper measurements and subjective visual crack detection. These existing practices often lead to inaccuracies and inconsistencies, making it difficult for laboratories and cement companies to evaluate compressive strength with precision. The proposed system integrates non-contact ultrasonic sensors for accurate cube dimension measurement and camera-based crack detection powered by image processing and machine learning, ensuring reliable and real-time monitoring of the testing process.

Furthermore, the system automatically calculates compressive strength by synchronizing force data with surface area measurements and crack initiation points, thereby eliminating manual errors and delays. Designed to be accurate, efficient, and user-friendly, the solution enhances the overall reliability of quality control in the construction industry. By leveraging modern technologies such as sensors, computer vision, and machine learning, this project provides a sustainable and innovative approach that supports safer, more efficient, and cost-effective construction practice

12.REFERENCES

[1] Kapadia, H. K., Patel, P. V., & Patel, J. B. (2023). Convolutional neural network based improved crack detection in concrete cubes. *International Journal of Computing and Digital Systems*, *13*(1), 341-352.

[2] Patil, M. S., Ghongade, R. B., & Salunke, R. V. (2023, October). Image processing and Machine learning in Concrete Cube Crack detection. In *2023 International Conference on New Frontiers in Communication, Automation, Management and Security (ICCAMS)* (Vol. 1, pp. 1-5). IEEE.

[3] Kassim, A. M., Jaafar, H. I., Azam, M. A., Abas, N., & Yasuno, T. (2013, August). Performances study of distance measurement sensor with different object materials and properties. In *2013 IEEE 3rd International Conference on System Engineering and Technology* (pp. 281-284). IEEE.

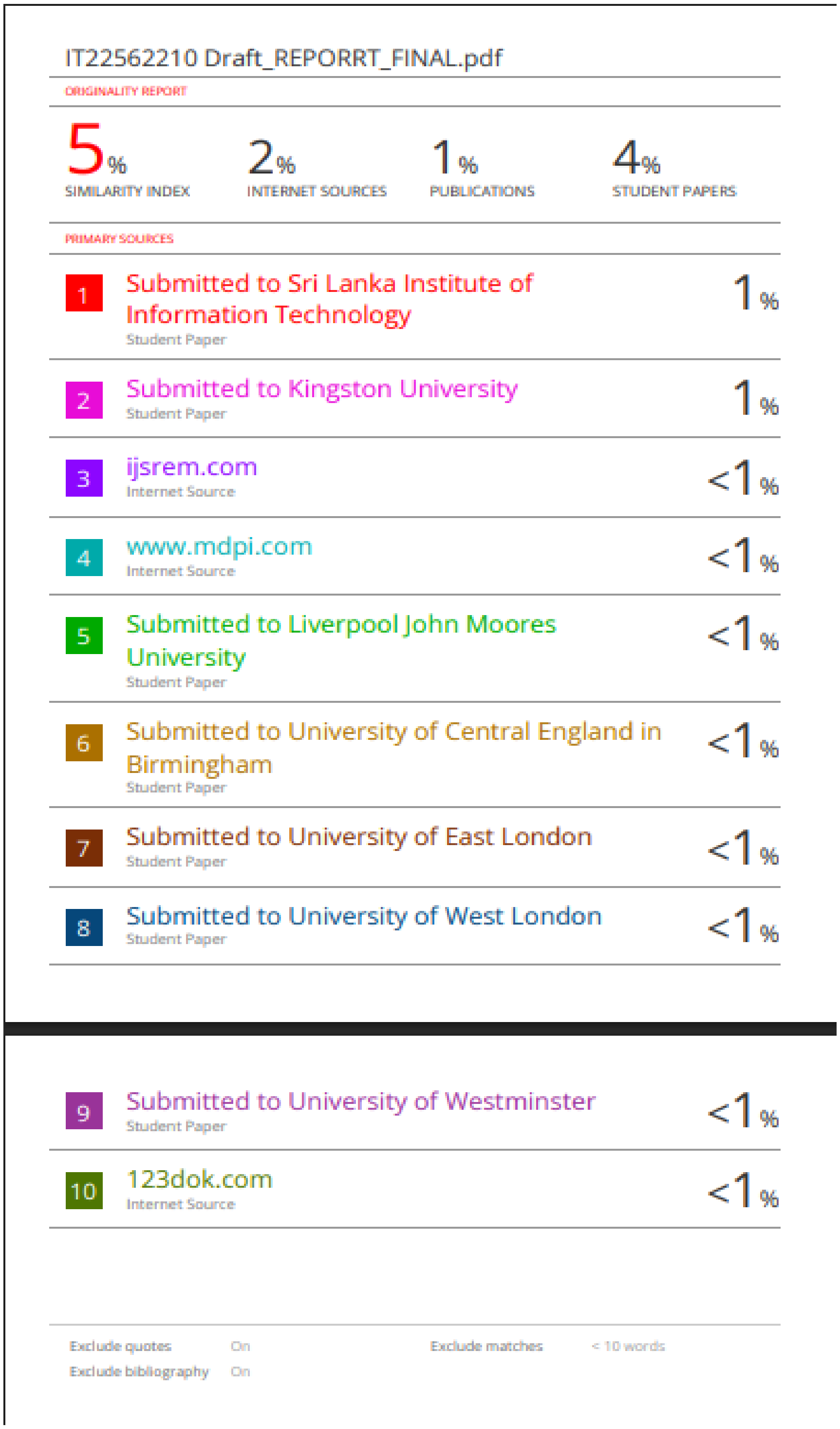
[4] Gkantou, M., Muradov, M., Kamaris, G. S., Hashim, K., Atherton, W., & Kot, P. (2019). Novel electromagnetic sensors embedded in reinforced concrete beams for crack detection. *Sensors*, *19*(23), 5175.

[5] Doğan, G., Özkiş, A., & Arslan, M. H. (2024). A new methodology based on artificial intelligence for estimating the compressive strength of concrete from surface images. *IngenIería e InvestIgacIón*, *44*(1), 4.

[6] Gond, a. (2024). Performance evaluation of iot based intelligent crack detection system for concrete building blocks.

[7] Gond, a. (2024). Performance evaluation of iot based intelligent crack detection system for concrete building blocks.

13.APPENDIX



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